

1

2

3

4

5

6

7

8

edge-rp2354a

Cart edge straight into the RP2354A (5 V-tolerant pads)

File: edge-rp2354a.kicad_sch

esp32s3-sd

ESP32-S3, microSD, status LED

File: esp32s3-sd.kicad_sch

usb-uart

USB-C, CP2102N bridge

File: usb-uart.kicad_sch

power

Power: console 5V / USB OR, 3.3V buck, RP LDO

File: power.kicad_sch

Fujiversal-Atari2600 Rev1 (RP2354A + ESP32-S3) – FujiNet for the Atari 2600

The cartridge port carries only A0–A12, D0–D7, +5V and GND (no R/W, clock or select: A12 is the decode). Every line goes straight from the edge to the RP2354A: GPIO2–14 = A0–A12, GPIO15–22 = D0–D7 (vcs_pins.h). GPIO0–25 are 5V-tolerant (FT) while IOVDD = 3.3V, and a 3.3V high clears the 6507’s TTL VIH (2.0V), so there are no buffers and no direction pin: the firmware’s per-access output enable is the only control. The whole RP runs from its own fast LDO (+3V3_RP) on the 5V rail, so IOVDD rises with the console’s 5V and the pads are powered before the bus can carry 5V into them. GPIO27 (ADC, not 5V-tolerant) senses the console 5V through 100k/150k; GPIO25 drives the activity LED; GPIO0/1 (debug UART) go to test pads. ESP32-S3-WROOM-1-N16R8 runs FujiNet (fujiversal-atari2600) and is the USB host of the RP (CDC, VID 0xCafe); S3 I04 / I05 pull RP RUN / QSPI_SS low through 1k, so the S3 can put the RP in BOOTSEL and re-flash it. RESET resets both chips; BOOTSEL + RESET enters the RP bootloader; first flash of the RP: SWD test pads. Power: console 5V through a P-FET (on only without USB) ORed with USB VBUS through a Schottky -> +5V -> 3.3V buck (S3, SD, bridge) and the RP LDO. USB-C: power + CP2102N UART for flashing the S3. microSD on SPI. Edge: 2x12 fingers, 2.54 mm pitch; pins 13–24 on the component face (console REAR), 1–12 on the label face.

Generated by tools/gen_sch.py

FujiNet

Sheet: /

File: Fujiversal-Atari2600-Rev1.kicad_sch

Title: **Fujiversal-Atari2600 Rev1**

Size: A3

Date: 2026-10-06

Rev: 1

KiCad E.D.A. 10.0.6

Id: 1/5







